THE STAGES OF A PRESENTATION OF AN AFFINE FORMAL ALGEBRAIC SPACE THICKEN THE SPACE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof displays one identification of a fibre product; it lies inside the counted span and is reproduced verbatim.

Editorial note. The counted spans carry 4 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The diagonal of an affine formal algebraic space is representable. Same chapter, Lemma lemma-diagonal-affine-formal-algebraic-space, invoked at proof lines 2066-2068. Gives representability of the maps from the stages.
- Presentations of an affine formal algebraic space and the Zariski local factorisation of a morphism through a stage. Same chapter, Definition definition-affine-formal-algebraic-space and the discussion following it, invoked at proof lines 2070-2074. Reduces the verification of the thickening property to a piece on which the morphism factors through a later stage.
- The transition morphisms of the presentation are thickenings, and thickenings are stable under base change. Same chapter, Definition definition-affine-formal-algebraic-space, in force at proof lines 2074-2077. Identifies the fibre product as a base change of a thickening.

Source reference. Original lemma 0AI8: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2010–2010. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original section 025U: sources/stacks-chow/tags.txt, source lines 2827–2827. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original section 02WI: sources/stacks-chow/tags.txt, source lines 3805–3805. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:formal-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original remark 0AIS: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2769–2769. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let S be a scheme. We say a sheaf X on $(Sch/S)_{fppf}$ is an *affine formal algebraic space* if there exist

(1) a directed set Λ,
(2) a system $(X_\lambda, f_{\lambda\mu})$ over Λ in $(Sch/S)_{fppf}$ where
   (a) each X_λ is affine,
   (b) each $f_{\lambda\mu} : X_\lambda \to X_\mu$ is a thickening,

such that

$$X \cong \operatorname{colim}_{\lambda\in\Lambda} X_\lambda$$

as fppf sheaves and X satisfies a set theoretic condition (see Remark 0AIS). A *morphism of affine formal algebraic spaces* over S is a map of sheaves.

Lemma 0.2. *Let $X_\lambda, \lambda \in \Lambda$ and $X = \operatorname{colim} X_\lambda$ be as in Definition 0.1. Then $X_\lambda \to X$ is representable and a thickening.*

Proof. The statement makes sense by the discussion in Spaces, Section 025U and 02WI. By Lemma 0AI8 the morphisms $X_\lambda \to X$ are representable. Given $U \to X$ where U is a scheme, then the discussion following Definition 0.1 shows that Zariski locally on U the morphism factors through some X_μ with $\lambda \leq \mu$. In this case $U \times_X X_\lambda = U \times_{X_\mu} X_\lambda$ so that $U \times_X X_\lambda \to U$ is a base change of the thickening $X_\lambda \to X_\mu$. □